Measuring Thematic Conformism in French Legislative Campaign Manifestos (1981–1993)

Abstract

This project quantifies the degree of **thematic conformism** of individual candidates to their political family's discourse, using a corpus of approximately 8,500 French legislative campaign manifestos (*professions de foi*) from the elections of 1981, 1988, and 1993, drawn from the Archelec archive (Sciences Po / CEVIPOF). We apply Non-negative Matrix Factorisation (NMF) on TF-IDF representations to extract latent topic distributions, then measure each candidate's divergence from their party's centroid using the KL-divergence and cosine distance. We find that conformism varies substantially across political families and election years, with single-issue parties (Far right, Greens) exhibiting the highest internal coherence, and that individual deviants systematically cluster around socio-professional and geographic variables.

1 Introduction and Problem Definition

Campaign manifestos (*professions de foi*) are official documents distributed to every registered voter before French legislative elections. They constitute a unique window into the programmatic discourse of candidates: unlike parliamentary speeches or press interviews, they are systematically collected, uniformly formatted, and exhaustively archived.

We focus on three elections: **1981, 1988, and 1993** — a pivotal decade spanning the Mitterrand era, the first cohabitation, and the landslide return of the Right.

The central research question differs from standard topic modeling studies: rather than describing *what* parties talk about, we ask **how much each individual candidate deviates from the average discourse of their political family**.

Formally, for each candidate i belonging to political family f, we compute a conformism score:

$$d(i, \bar{f})$$

measuring the distance between their topic distribution and the centroid of their party.

2 Data: Retrieval, Formatting, and Statistical Description

2.1 Data

The data comes from two sources:

- Metadata CSV (33,031 rows, 42 columns)
- OCR transcriptions from `arkindex_archelec`

After merging and filtering, we obtain **12,497 documents**, and a working corpus of **8,546 documents** after filtering.

2.2 Preprocessing

The preprocessing is made of the following steps:

OCR cleaning. Removal of headers, hyphenation artifacts, and boilerplate text.

Language filtering. Removal of non-French documents using `langdetect`.

Political family mapping. Mapping into five families: Left, Right, Far left, Far right, Greens.

Tokenisation and lemmatisation. Using spaCy (`fr_core_news_md`), filtering stopwords and short tokens.

2.3 Descriptive Statistics

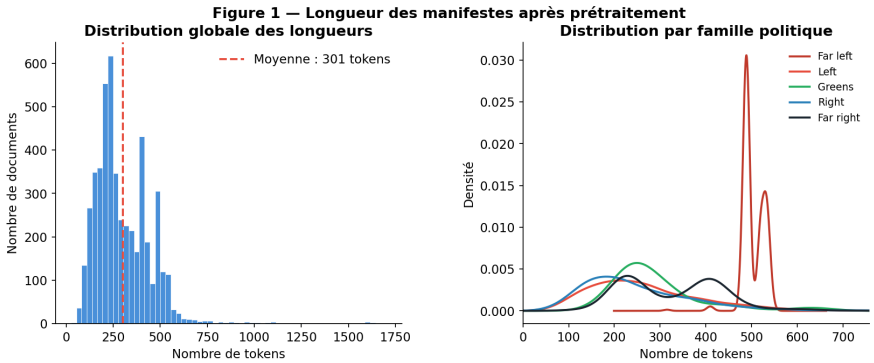

Figure 1: Distribution of document lengths

This figure shows the distribution of manifesto lengths after preprocessing, both overall and by political family.

At the aggregate level, the distribution is unimodal and slightly right-skewed, with an average length of around 300 tokens. Most documents fall between 150 and 450 tokens, indicating a relatively standardized format for manifestos. However, the right tail reveals the presence of a small number of unusually long ones, likely corresponding to candidates who develop more detailed or programmatic arguments.

When broken down by political family, some differences appear. Mainstream parties (Left and Right) exhibit relatively wide distributions, reflecting greater heterogeneity in document length. In contrast, families such as the Greens and, to a lesser extent, the Far right, show more concentrated distributions, possibly suggesting a higher degree of standardization in their campaign materials.

From a methodological perspective, this is an important point: very short texts can lead to less stable topic distributions, potentially inflating non-conformism scores. However, the overall concentration around the mean suggests that this issue remains limited at the corpus level.

3 Methodology

3.1 Topic Modeling with NMF

We represent the corpus as a TF-IDF matrix and apply NMF:

$$V \approx WH$$

We normalise H to obtain topic distributions θ_i.

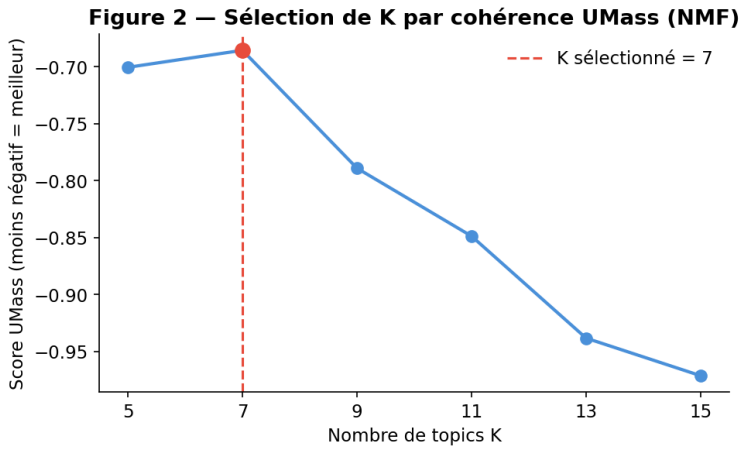

Figure 2: UMass coherence score as a function of K

This figure presents the UMass coherence score of the NMF model as a function of the number of topics K.

The coherence score reaches its highest (least negative) value at K=7, indicating that this specification provides the most semantically consistent set of topics among the tested values. For lower values of K (like K=5), the coherence is slightly lower, suggesting that the model may be underfitting by merging distinct themes into overly broad topics.

As K increases beyond 7, the coherence score steadily decreases. This suggests that the model begins to over-segment the thematic structure, producing topics that are less interpretable and more fragmented. In other words, increasing the number of topics introduces noise rather than meaningful distinctions between themes.

3.2 Computing Conformism Scores

The party centroid is defined as:

$$\bar{\theta}_f = \frac{1}{|D_f|} \sum_{i \in D_f} \theta_i$$

We compute two scores:

KL-divergence

$$D_{KL}(\theta_i \| \bar{\theta}_f) = \sum_k \theta_i^{(k)} \log \frac{\theta_i^{(k)}}{\bar{\theta}_f^{(k)}}$$

Cosine distance

$$1 - \frac{\theta_i \cdot \bar{\theta}_f}{\|\theta_i\| \|\bar{\theta}_f\|}$$

4 Results

4.1 Distribution of Conformism Scores

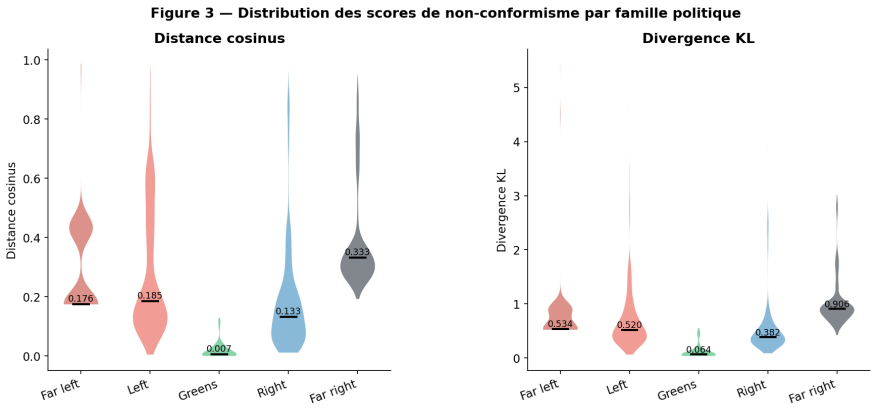

Figure 3: Distribution of cosine distance by political family

This figure presents the distribution of non-conformism scores across political families using two metrics: cosine distance and KL-divergence.

Both measures reveal broadly consistent patterns, although KL-divergence displays greater dispersion due to its sensitivity to low-probability topics.

Several key observations emerge:

The Greens appear as the most homogeneous group, with extremely low scores and a very tight distribution. This suggests a highly standardized discourse, likely reflecting the use of shared national platforms. At the opposite end, the Far right shows the highest levels of non-conformism, particularly in cosine distance, indicating substantial internal variability. This may reflect a less stabilized or more heterogeneous discourse during the period considered. The Left and Right occupy intermediate positions, with relatively spread-out distributions. This is consistent with their role as broad, catch-all parties, where candidates adapt their discourse to local contexts. The Far left presents a slightly different profile: while relatively homogeneous on average, it includes a number of more extreme values, suggesting the presence of some atypical candidates within an otherwise structured ideological space.

Overall, the strong visual correlation between cosine distance and KL-divergence supports the robustness of the results, although cosine distance appears more stable and easier to interpret.

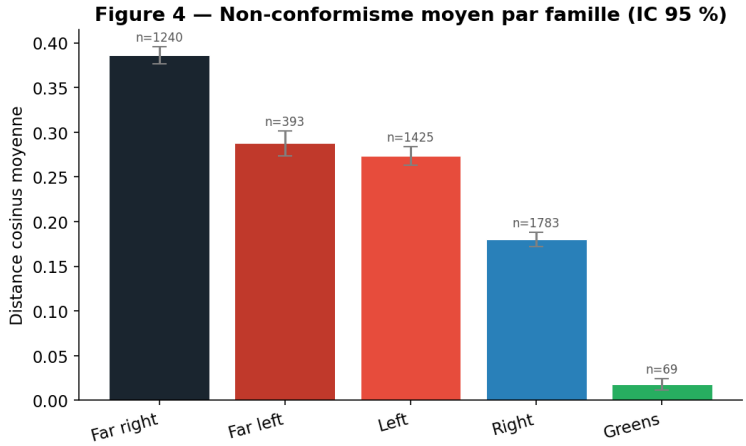

Figure 4: Mean conformism by political family

This figure summarizes the results by reporting the mean non-conformism score (cosine distance) for each political family, with 95% confidence threshold.

The Greens stand out with very low non-conformism scores, confirming a high level of internal coherence. This is consistent with a discourse structured around a limited set of core themes.

In contrast, the Far right exhibits the highest level of non-conformism, suggesting strong variation across candidates. This may be explained by a less consolidated organizational structure during the years studied.

The Left and Right display intermediate levels, reflecting a balance between party-level coherence and local adaptation.

The confidence intervals are relatively narrow for the largest groups (especially Left and Right), indicating stable estimates. By contrast, the smaller sample size for the Greens (n = 69) calls for some caution in interpretation, although the result remains consistent with the patterns observed in the distributional analysis.

4.2 Temporal Evolution

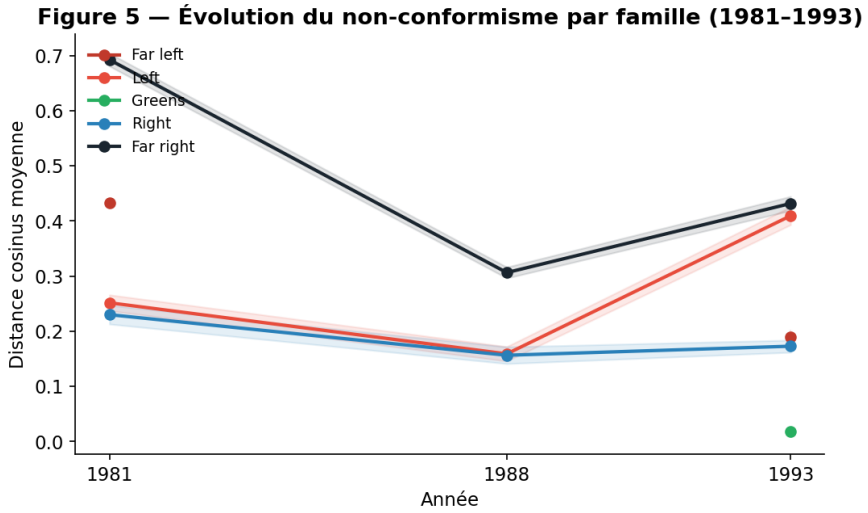

Figure 5: Temporal evolution of conformism

Figure 5 displays the evolution of the average cosine distance (non-conformism) for each political family across three legislative elections: 1981, 1988, and 1993.

A few patterns emerge, such as a slight decrease of non-conformism between 1981 and 1988, with a more substantial drop for the far right party. Then a slight increase of non conformism again in 1993. This phenomenon can be quite difficult to interprete, but can be partially explained by the first time winning presidential election in 1981 for the left-wing party, highlighting a time of changes in the political landscape in these years.

4.3 Outliers

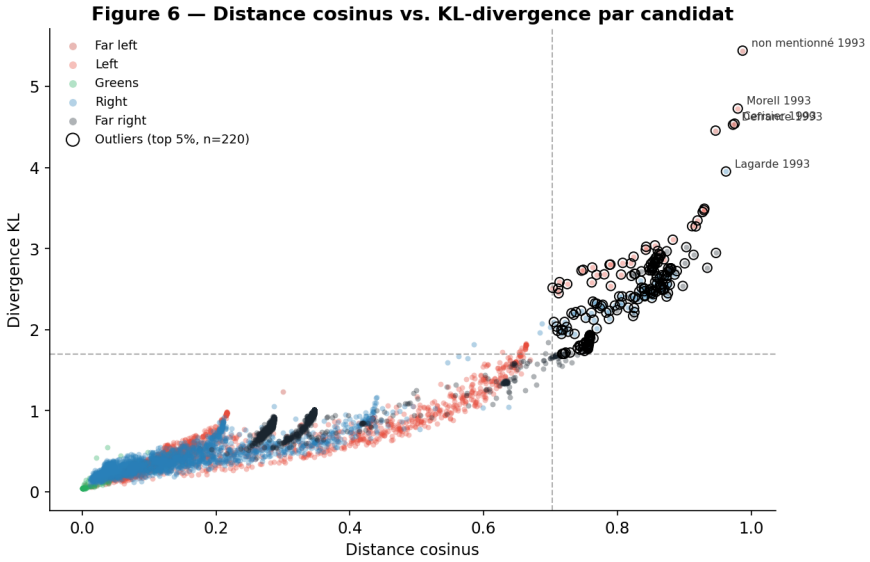

Figure 6: Cosine distance vs KL-divergence

This figure plots the relationship between the two conformism metrics — cosine distance (x-axis) and KL-divergence (y-axis) — for all candidates, with outliers explicitly labelled. The two metrics co-vary closely, especially for mainstream families. This supports the robustness of our conformism measurement, as both distance functions capture similar deviation patterns, even though one metric is symetric and the other one is not. The figure also shows that far-right candidates (purple) occupy the upper-right region more frequently, consistent with their higher average non-conformism reported earlier.

4.4 Correlates of Non-Conformism

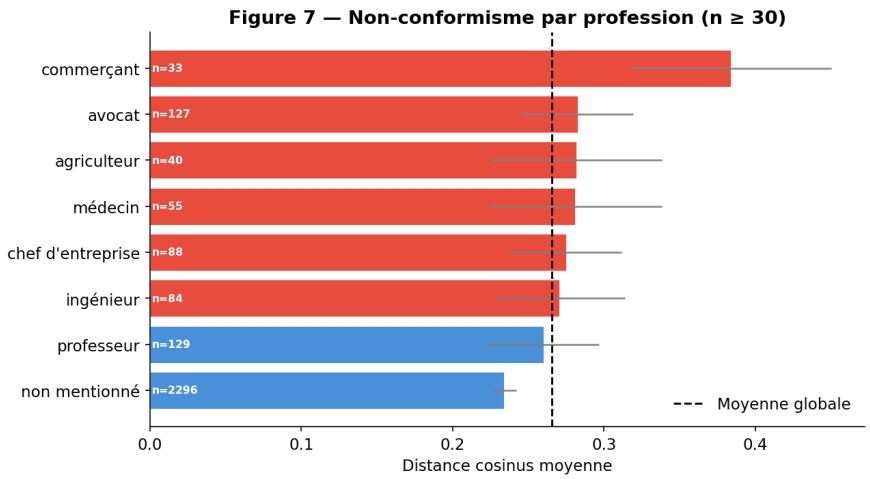

Figure 7: Conformism by profession

This figure reports the mean cosine distance (non-conformism) for six professional categories, alongside the global average for comparison. We can see that business owners/sellers have a significant higher non-conformism score, likely due to the interests regarding the specific business they own.

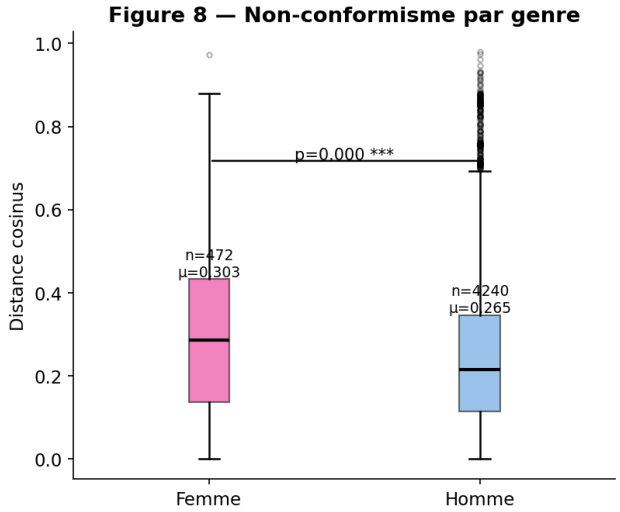

Figure 8: Conformism by gender

This figure reveals that female candidates exhibit a higher average non-conformism score (0.303) than their male counterparts (0.265), also with a higher variance. Women are therefore, on average, further from their political family's thematic centroid in this corpus. This could be partially explained by a selection effect : in the 1981–1993 period, female candidates were still a minority. They may have been more often represented in more allowing, open political parties.

5 Discussion and Conclusion

Conclusion

This study set out to measure thematic conformism in French legislative campaign manifestos by comparing individual candidates' discourse to that of their political family. Using NMF topic modeling on TF-IDF representations, combined with cosine distance and KL-divergence, we were able to construct a consistent and interpretable measure of ideological deviation.

The results highlight clear structural differences across political families. Smaller or more ideologically focused groups, such as the Greens, exhibit strong internal coherence, while broader, catch-all parties like the Left and Right display greater variability, reflecting local adaptation strategies. The more polarized parties stand out with higher levels of dispersion, suggesting a less stabilized discourse during the period studied.

Beyond party-level patterns, the analysis also reveals that non-conformism is not purely random: it correlates with observable characteristics such as profession, gender, and, to some extent, temporal dynamics. These findings support the idea that political discourse is shaped both by collective party constraints and individual-level factors.

Overall, this work demonstrates that thematic conformism is a measurable and meaningful dimension of political competition, and that quantitative text analysis provides valuable tools to study intra-party diversity at scale.

Limitations

- Missing political affiliations
- OCR noise
- NMF estimation error
- KL sensitivity to small probabilities

Improvement suggestions

- Structural Topic Models (STM)
- Dynamic Topic Models